

NAME: _____

PERIOD: _____

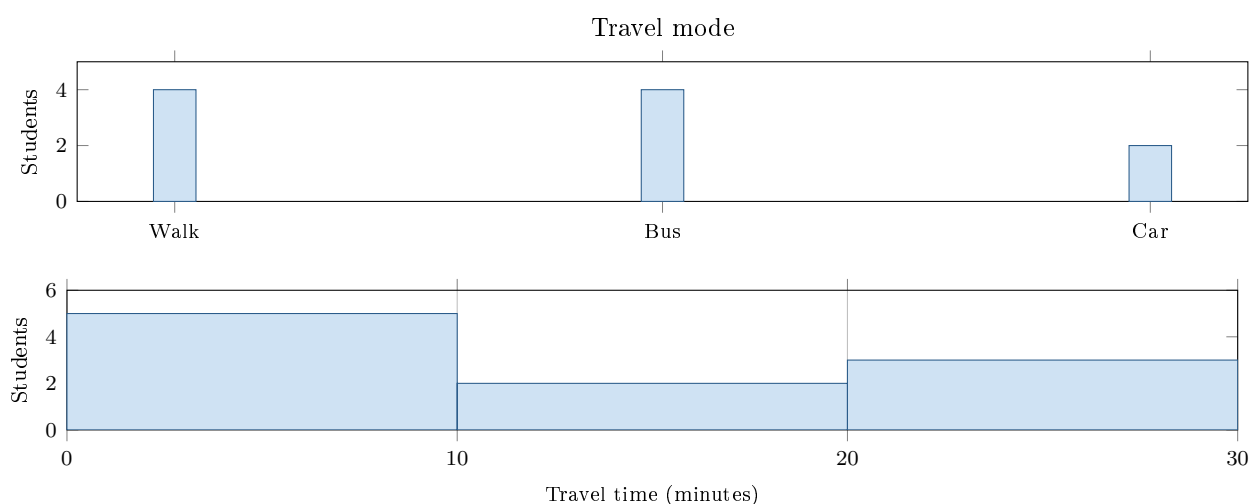
One Commute, Two Questions

AP Statistics · Unit 1 · Topics 1.4–1.5 (CED 1.4–1.5)

[STAR] THE PROBLEM

Suppose ten students report their usual travel mode and travel time. Both graphs summarize these same students; the individual mode–time links are not shown. Histogram intervals include the left endpoint only.

Ari: “Bus is the most common mode, so the bus riders must be the students with long trips. ” **Dev:** “The histogram answers every question about these commutes; we can drop the mode graph.”



FIRST TAKE — one sentence before you work the parts

Which report would you send to the principal, or would you revise both? Name one visible feature.

[BOOK] TWO TOPICS, ONE DATA STORY

1.4: Bar charts compare categorical counts or proportions; a zero baseline keeps heights proportional.

1.5: Histograms group quantitative values into ordered intervals; bin width affects detail.

Neither pooled graph preserves the link between each student's mode and travel time.

Work (a)–(c).

DOK 1 (a) Name the most common travel mode or modes and their counts.

DOK 2 (b) How many students have trips of at least 10 minutes? Explain how the histogram bars answer this and why they do not give each exact travel time.

DOK 3 (c) ★ In three to five sentences, judge both reports. Choose a display for comparing travel modes and a display for describing travel times. Give one corrected numerical claim and explain what extra information is needed to connect long trips with bus travel.

Frame: I would report _____ because _____; the display cannot establish _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.